Attendance Calculator

Automated Attendance Tracking and Prediction System

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Problem & Motivation

The Challenge

Mandatory Attendance

75% required per subject for exams.

Manual Tracking Issues

Time-consuming (15-20 mins/week) and error-prone.

Lack of Foresight

Students realize low attendance too late.

Complex Calculations

Difficulty with holidays and multiple subjects.

Why we Built this?

Stop the Panic

Provide easy, accurate, and instant updates.

Smart Planning

Change attendance management to proactive planning.

Key Questions Answered

"How many classes can I skip?" "What is the highest percentage I can reach?"

Proposed Solution & Objectives

A web-based **Attendance Calculator** automates tracking and offers predictive analysis.

Automate Tracking

Accept weekly timetables and track current status.

Predict Future Scenarios

Calculate outcomes for "Attend All," "Bunk All," or "Selective Bunking."

Goal-Oriented Tools

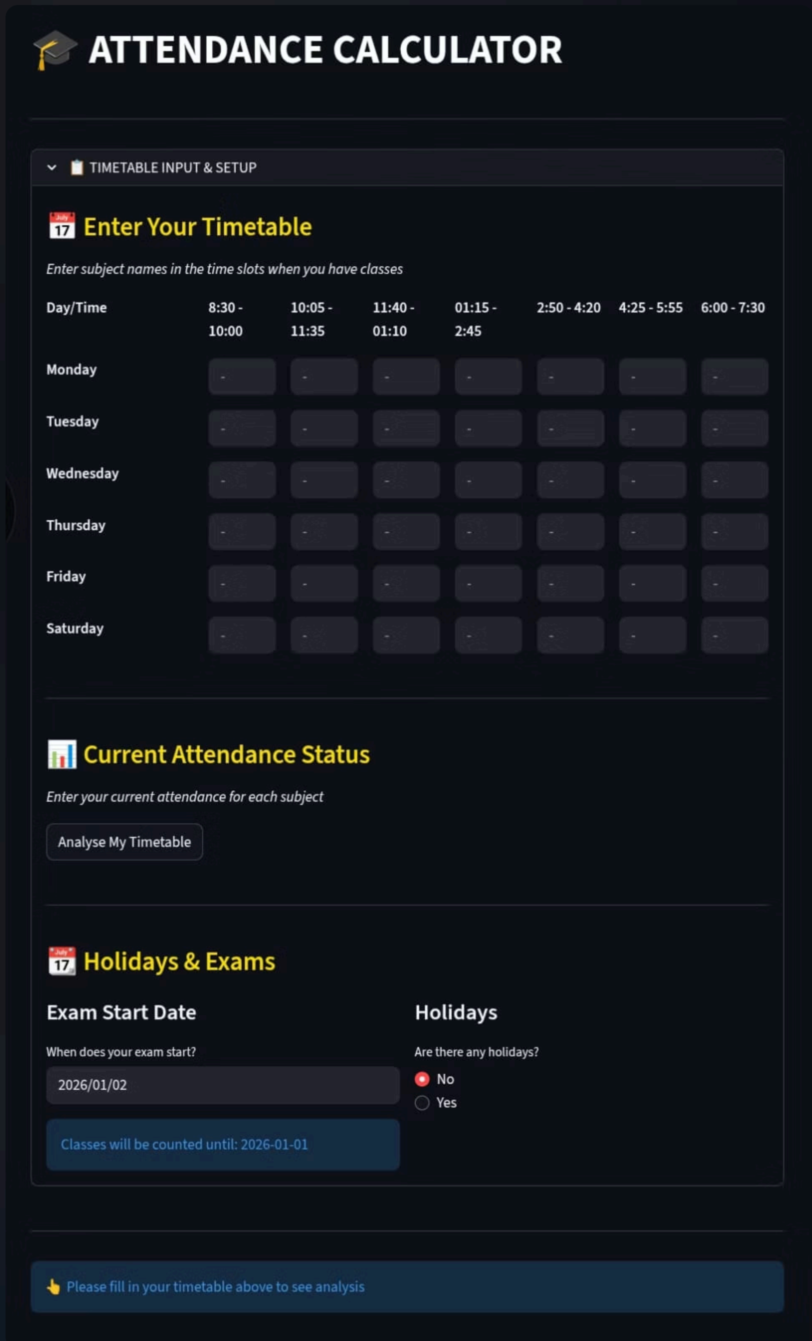
Calculators for maximum bunkable classes and reaching target percentages.

Holiday Integration

Accurate remaining class counts by excluding holidays.

Real-Time Feedback

Instant, color-coded visual analysis.



System Architecture: Three-Tier Design

1

Input Layer

- User Timetable Grid
- Current Attendance Data
- Exam Dates & Holiday Periods

2

Processing Layer

- Parsing: Extracts subjects
- Calculation: Computes % & remaining classes
- Prediction: Runs algorithms

3

Output Layer

- Color-coded Status Cards
- Scenario Predictions
- Next 5 Classes Projection

Subject-Wise Detailed Analysis

Real-Time Status

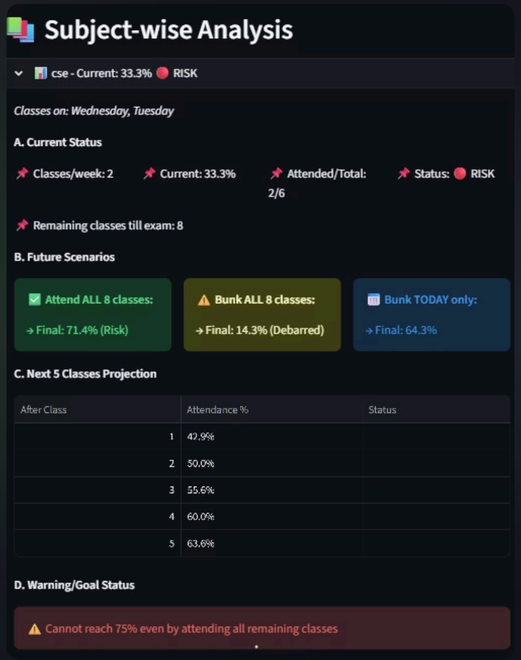
Displays Current Percentage, Attended/Total Ratio, and a color-coded 'SAFE/RISK' indicator. Also counts exactly how many classes remain until the exam.

Predictive Scenarios

Calculates three specific outcomes: Attend All (Max Potential), Bunk All (Worst Case), and Bunk Today (Immediate Impact).

Next 5 Classes Trend

A step-by-step projection showing exactly how your percentage changes after each of the next 5 classes, helping you spot safe bunking windows.



Solving the Holiday and Remaining Class Challenge

Students cannot accurately determine future attendance because they struggle to subtract holidays and match variable weekly schedules.

The Solution: Iterative Class Counting

1

Start Date to Exam Date

The system iterates through every single day from today until the exam date.

2

Holiday Exclusion

Checks if the date falls within a user-defined holiday period; if YES, it skips the day.

3

Schedule Match

If it's a regular day, it checks the subject's schedule (e.g., Is today a Monday?).

4

Count & Tally

Only classes scheduled on non-holiday weekdays are added to the total remaining count.

Technology Stack

Python 3.x

- Selected for its rich built-in `datetime` **library**, simplifying complex date arithmetic and holiday exclusion.
- **Readable syntax** supports rapid prototyping and maintains clear logic for the core predictive algorithms.

Streamlit

- Enables **rapid development** of interactive web applications using pure Python.
- Eliminates the need for writing complex **HTML, CSS, or JavaScript.**
- Ideal for data-centric tools due to its reactive nature and built-in widgets.

Pandas

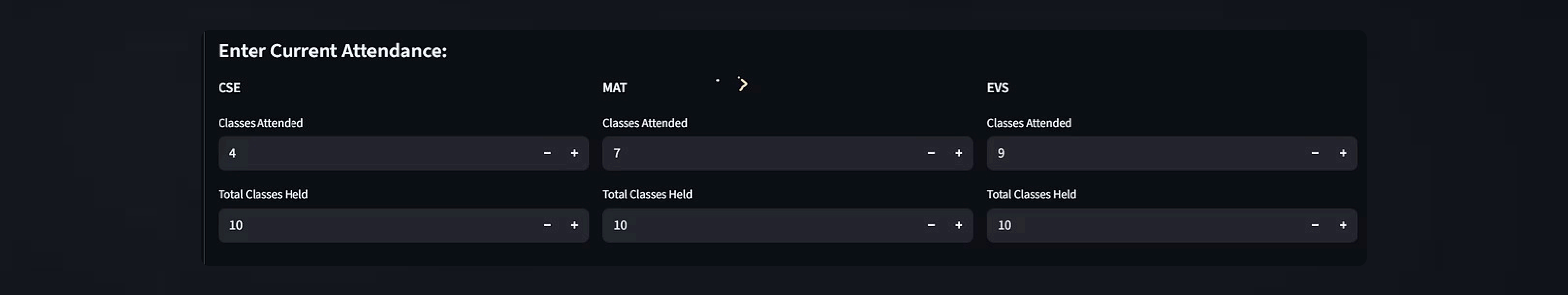
- Used for efficient handling and structuring of the **7-day by 7-slot tabular timetable data.**
- Facilitates quick extraction and counting of classes per subject across the week

Datetime

- Crucial for implementing Algorithm 2 to **iterate through dates, exclude holidays,** and accurately count remaining classes until the exam date.
- Simplifies the arithmetic needed for projection and goal calculations

JSON

- Used within Streamlit's **Session State** to store and retrieve the user's complete timetable and attendance data across different page interactions.
- Ensures the application remains **stateless** between calculations without requiring a database.



Key Features

1

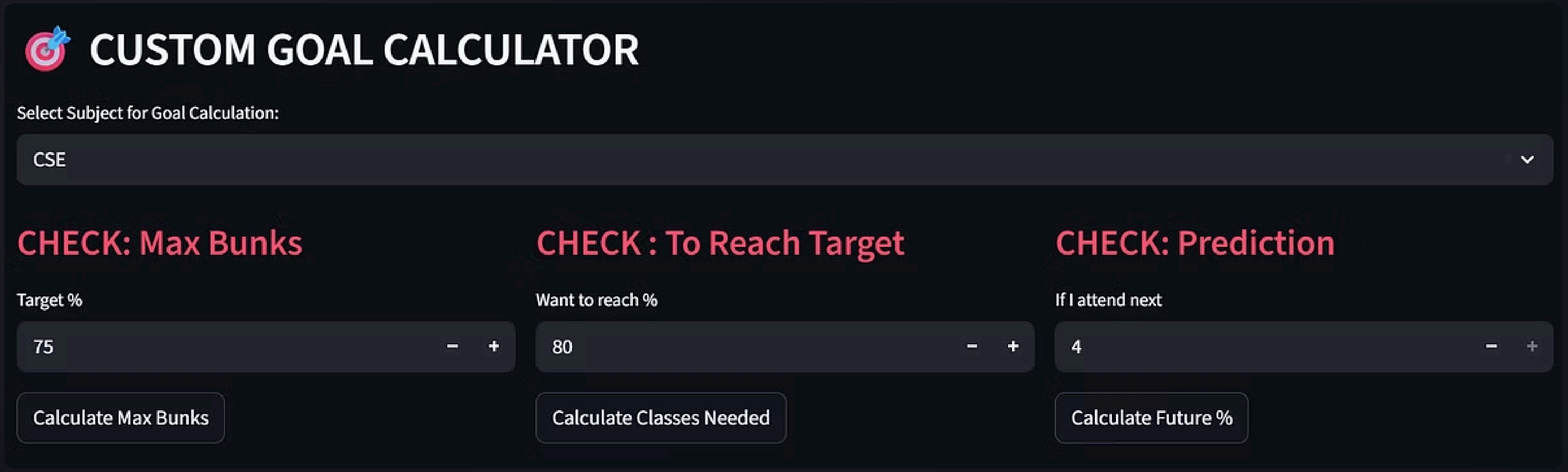
Smart Timetable Grid

Dynamic parsing of a 7-day × 7-slot grid. Automatically extracts unique subjects and their weekly frequency.

2

Predictive Analysis Cards

- Current Status: Real-time percentage display.
- Future Scenarios: Instantly shows impact of attending vs. missing classes.
- Next 5 Classes: Projects percentage changes after each of the next 5 specific classes.



Goal Calculators

Max Bunks Calculator

"How many more classes can I safely miss and still stay above 75%?"

Target Reach Calculator

"I have 65%. How many classes must I attend to reach 80%?"

Prediction Calculator

"If I attend the next 5 classes, what will my percentage be?"

Visual Alerts & Status System

The calculator provides instant, color-coded feedback for every subject.

🚦 Quick Status Indicators

🟢 **SAFE**	75% and above	Comfortable buffer.
🟡 **WARNING**	60-74%	Start attending classes carefully.
🔴 **RISK**	Below 60%	Immediate action required.

🔔 Critical Alerts

Urgent Recovery Alert

If below 75%, system shows exact classes to attend to regain SAFE status.

Impossible Goal Warning

Notifies if target is impossible, displays maximum achievable percentage.

Final Percentage Determination

How the system determines the final percentage for different future actions.

Attend All	$\frac{\text{Attended}_{\text{Current}} + \text{Remaining}_{\text{Total}}}{\text{Total}_{\text{Current}} + \text{Remaining}_{\text{Total}}} \times 100$	"If you attend all 20 classes, your attendance will be 78.6%."
Bunk All	$\frac{\text{Attended}_{\text{Current}}}{\text{Total}_{\text{Current}} + \text{Remaining}_{\text{Total}}} \times 100$	"If you bunk all 20 classes, your attendance will drop to 50%."
Max Bunks	Iteratively test bunk counts (N=0, 1, 2...) until the percentage drops below the user's target.	"You can safely miss 7 more classes to maintain 75%."